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# A new basis set method for quantum scattering calculations

William H. Miller and Bernadette M. D. D. Jansen op de Haar
*Department of Chemistry, University of California, and Materials and Molecular Research Division, Lawrence Berkeley Laboratory, Berkeley, California 94720*

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A new basis set approach for quantum scattering calculations is described and tested on model problems of elastic and inelastic collisions. The approach is essentially the Kohn variational method, but applied to the $S$ or $T$ matrix directly rather than to the $K$ matrix as is normally done; it is seen that the result of the present approach is not equivalent to the usual Kohn method (i.e., for the $K$ matrix) and is indeed preferable to it. The present approach is seen to have the same structure as the complex scaling/coordinate rotation expressions for the $T$ matrix (but with some added features). Its potential advantage over the Schwinger variational method, another useful basis set technique, is that matrix elements of the Green's function for some reference Hamiltonian are not required; the present method requires only matrix elements of the Hamiltonian itself between the basis functions. The essential reason for all of these desirable features is that the basis set which is used incorporates the correct scattering boundary conditions.

## I. INTRODUCTION

The purpose of this paper is to describe a new approach for carrying out quantum scattering calculations. It is essentially a Kohn-like variational method,[1,2] but applied to the $T$ matrix (or equivalently to the $S$ matrix) rather than to the $K$ matrix, as is usually done. It will be see that this approach is not equivalent to the usual Kohn method (for the $K$ matrix), and in fact is superior to it in having none of the spurious singularities that plague the usual method.[3] It will also be seen that the method bears an interesting and illuminating relation to the complex scaling method of Nuttall,[4] Reinhardt,[5] and McCurdy and Rescigno.[6,7]

It is important that the approach, described in Secs. II and III, is applicable to both local and nonlocal interaction potentials, for various propagator methods—e.g., renormalized Numerov, log derivative,[8] $R$-matrix propagator,[9] etc.—already do a very good job for purely local interaction potentials. The specific applications that are of interest to us are chemically reactive collisions, but the approach should also be useful for electron–atom/molecule scattering. Both of these processes involve nonlocal interaction potentials.

We find it most useful to think do the method as providing an appproximation to the Green's function $G^+(E)$,

$$G^+(E) \equiv \lim_{\epsilon\to 0}(E + i\epsilon - H)^{-1}, \tag{1.1}$$

where $H$ is the total Hamiltonian of the system. Scattering amplitudes, or $S$-matrix elements, are then proportional to matrix elements of the $T$ operator

$$T(E) = V + VG^+(E)V. \tag{1.2}$$

The approximation (*vide infra*) for $G^+(E)$ has the generic form

$$G^+(E) \simeq \sum_{l,l'=0}^{N} |u_l\rangle(\langle u_l|E - H|u_{l'}\rangle)^{-1}\langle u_{l'}|, \tag{1.3}$$

where $\{u_l\}$ is some basis set in the space of the Hamiltonian, and $(\langle u_l|E - H|u_{l'}\rangle)^{-1}$ is shorthand for the $l,l'$ element of the inverse of the $(N+1)\times(N+1)$ matrix $\langle u_l|E - H|u_{l'}\rangle$. The cognizant reader will immediately recall that over the years there have been several basis-set approximations for quantum scattering that involve expressions like Eq. (1.3) [together with Eq. (1.2)]. The attraction of such approaches is that the scattering calculation is reduced to bound state-like quantum calculations, i.e., to computing matrix elements of the Hamiltonian in a basis set and then performing linear algebra. It is well recognized, however, that Eq. (1.3) cannot apply in its most naive interpretation—e.g., with a real set of square-integrable basis functions—because for finite $N$ the real symmetric matrix $\langle u_l|H|u_{l'}\rangle$ would have a discrete set of real eigenvalues, so that as a function of $E$, $G^+(E)$ from Eq. (1.3) would have a series of poles rather than being a smooth function of $E$ (as it should be for $E$ in the scattering region, i.e., in the continuous spectrum of $H$). Several analytic continuation methods[10,11] have been described for overcoming this pathology, and though they can indeed work in some cases, they have not led to generally useful methods.

A more recent variant of these approaches is based on the coordinate rotation/complex scaling ideas of Nuttall,[4] Reinhardt,[5] and McCurdy and Rescigno.[6] (In fact, it was our interest in this approach that led us to the work presented in this paper.) In the particular version used by McCurdy and Rescigno,[6] at least some of the basis functions $\{u_l\}$ need to be complex, and the bra states $\langle u_l|$ in Eq. (1.3) are replaced by $\langle u_l^*|$, where $*$ denotes complex conjugation. (Or equivalently, one does *not* complex conjugate the wave function on the left in the integrals for the matrix elements; i.e.,

$$\langle u_l^*|H - E|u_{l'}\rangle = \int d\tau\, u_l(H - E)u_{l'}. \tag{1.4}$$

Since $\langle u_l^*|H|u_{l'}\rangle$ is a complex-symmetric matrix, it has no real eigenvalues and thus Eq. (1.3) has no unphysical singularities for real $E$. This method has been seen to work in some cases, but in general the stability of the results with respect to the complex rotation angle is somewhat tenuous, particularly so if the wave vector $k$ is large, as is the case with heavy particle (as opposed to electron) scattering.



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The approach we use, as presented in Secs. II and III, has the same general form as Eqs. (1.3) and (1.4), yet there are no complex scaling parameters, no anomalous polar singularities, and no analytic continuation is involved. Furthermore, only a modest number of basis functions is required, and the results (e.g., in Sec. IV) are stable with respect to increasing the size of the basis set and with respect to variation of nonlinear parameters in the basis. The essential reason for all these desirable characteristics is that the basis we use incorporates the *correct boundary conditions*.

Finally, one should make note of the Schwinger variational method,[12] which is also a basis set method, though not of the same form as discussed above. The Schwinger method, which is essentially a basis set expansion of the interaction potential, has been used extensively for electron—atom/molecule scattering,[13] and our group has shown that it also works well for chemically reactive scattering.[14] It has all the "nice" characteristics noted in the preceding paragraph, but a major drawback is that it requires the construction of the Green's formalism $G_0$ for some reference Hamiltonian $H_0$. To be efficient, especially for the case of reactive scattering, it is important *not* to choose $H_0$ as the free particle Hamiltonian but rather as a distorted wave reference. The construction of, and subsequent manipulations with $G_0$ then turn out to be the most time consuming part of our reactive scattering calculations,[14] and this is the main reason we have been looking for a way to avoid it. The approach which is the subject of this paper has essentially all the desirable properties of the Schwinger variational method, and it requires only matrix elements of the Hamiltonian, Eq. (1.3), and not those of $VG_0V$.

Section II first describes the approach for the simple case of potential scattering, and its generalization of multichannel scattering is given in Sec. III. Some numerical applications to elastic and to multichannel scattering are presented in Sec. IV.

## II. A VARIATIONAL EXPRESSION FOR $G^+(E)$

The discussion in this section will be restricted to $s$-wave potential scattering in order to focus on the essential ideas. The generalization to inelastic and reactive scattering is made in the next section.

Consider a generic matrix element of the Green's function,

$$\langle f|G^+(E)|g\rangle \equiv \lim_{\epsilon\to 0}\langle f|(E+i\epsilon-H)^{-1}|g\rangle, \tag{2.1}$$

where $\langle f|$ and $|g\rangle$ are arbitrary states. [It is necessary only that the wave functions $f(r)$ and $g(r)$ go to zero faster than $1/r$ as $r\to\infty$.] The functional $I$,

$$I[\chi,\tilde{\chi}] \equiv \langle f|\chi\rangle + \langle\tilde{\chi}|g\rangle + \langle\tilde{\chi}|H-E|\chi\rangle, \tag{2.2}$$

is a variational expression for $\langle f|G^+|g\rangle$, where $|\chi\rangle$ and $\langle\tilde{\chi}|$ are independent "trial functions." To see this, note that the formal variation of $I$ with respect to $\langle\tilde{\chi}|$ is

$$\delta I = \langle\delta\tilde{\chi}|g\rangle + \langle\delta\tilde{\chi}|H-E|\chi\rangle, \tag{2.3}$$

so that $\delta I = 0$ implies that

$$(H-E)|\chi\rangle + |g\rangle = 0; \tag{2.4}$$

the exact function $|\chi\rangle$ which extremizes $I$ is thus

$$|\chi\rangle = G^+(E)|g\rangle. \tag{2.5}$$

[It will be verified below that $G^+(E)$ is the correct inverse operator to $(E-H)$.] Similarly, a formal variation with respect to $|\chi\rangle$ and the requirement $\delta I = 0$ leads to the following solution for $\langle\tilde{\chi}|$:

$$\langle\tilde{\chi}| = \langle f|G^+(E). \tag{2.6}$$

Equations (2.5) and (2.6) give the formally exact functions $|\chi\rangle$ and $\langle\tilde{\chi}|$ which extremize the functional $I$ of Eq. (2.2); if these solutions for $|\chi\rangle$ and $\langle\tilde{\chi}|$ are substituted into Eq. (2.2), then

$$I = \langle f|G^+|g\rangle,$$

demonstrating that the functional $I$, Eq. (2.2), is indeed a variational expression for the Green's function matrix element, Eq. (2.1).

It is, of course, not possible to use Eqs. (2.5) and (2.6) directly, but they are important in showing the *boundary conditions* which the exact functions $|\chi\rangle$ and $\langle\tilde{\chi}|$ satisfy.[15] Thus the wave function $\chi(r)\equiv\langle r|\chi\rangle$ is regular at $r=0$ [i.e., $\chi(0)=0$], and for large $r$,

$$\chi(r)\sim e^{ikr}\times\text{const}, \tag{2.7}$$

where $k=\sqrt{2mE/\hbar^2}$. Similarly, Eq. (2.6) shows that

$$|\tilde{\chi}\rangle = G^-(E)|f\rangle, \tag{2.6'}$$

so that the wave function $\tilde{\chi}(r)\equiv\langle r|\tilde{\chi}\rangle$ is also regular at $r=0$ and for large $r$,

$$\tilde{\chi}(r)\sim e^{-ikr}\times\text{const}. \tag{2.8}$$

We thus choose a linear variational trial function for $\chi(r)$,

$$\chi(r) = \sum_{l=0}^{N} u_l(r)c_l, \tag{2.9}$$

where $u_0(r)$ is chosen to impose the boundary condition (2.7),

$$u_0(0) = 0,$$

$$\lim_{r\to\infty} u_0(r)\sim e^{ikr}.$$

Several possible choices for $u_0(r)$, are

$$u_0(r) = e^{ikr}(1-e^{-\alpha r}) \tag{2.10a}$$

$$= e^{ikr} - e^{-\alpha r} \tag{2.10b}$$

$$= \cos(kr)(1-e^{-\alpha r}) + i\sin(kr). \tag{2.10c}$$

The functions $\{u_l(r)\}$, $l=1,\ldots,N$ are any convenient square integrable set and are assumed (without restriction) to be real. Similarly, a linear variational trial for $\tilde{\chi}(r)$ which satisfies the boundary condition equation (2.8) is

$$\langle r|\tilde{\chi}\rangle \equiv \tilde{\chi}(r) = \sum_{l=0}^{N} u_l(r)^* b_l, \tag{2.11}$$

where $*$ denotes complex conjugation and $\{u_l(r)\}$, $l=0,\ldots,N$ is the same basis as for the expansion of $\chi(r)$.

Substituting the linear trial forms, Eq. (2.9) and (2.11), into the functional $I$, Eq. (2.2), and varying the coefficients $\{c_l\}$ and $\{b_l\}$ to extremize the functional leads to the following result for the Green's function matrix element:



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$$\langle f|G^+(E)|g\rangle = \sum_{l,l'=0}^{N} \langle f|u_l\rangle(\langle u_l^*|E-H|u_{l'}\rangle)^{-1}\langle u_{l'}^*|g\rangle, \quad (2.12)$$

which is the basic result of the paper. [As noted before, $(\langle u_l^*|\cdots|u_{l'}\rangle)^{-1}$ denotes the $l,l'$ element of the inverse of the matrix $\langle u_l^*|\cdots|u_{l'}\rangle$.] We note that Eq. (2.12) applies as is even if the functions $\{u_l\}$ are not normalized and not orthogonal.

At this point it is useful to verify that the choice of boundary condition is correct by considering the coordinate representative of $G^+(E)$ from Eq. (2.12), i.e.,

$$\langle r|G^+(E)|r'\rangle = \sum_{l,l'=0}^{N} u_l(r)(\langle u_l^*|E-H|u_{l'}\rangle)^{-1}u_{l'}(r') \quad (2.12')$$

(since $\langle u_l^*|r'\rangle = \langle r'|u_{l'}\rangle$). For $r\to\infty$ and $r'$ finite, only the $l=0$ term survives in Eq. (2.12′), so that

$$\lim_{r\to\infty}\langle r|G^+(E)|r'\rangle = e^{ikr}\times\text{function}(r'). \quad (2.13a)$$

Similarly, for $r'\to\infty$ and $r$ finite, only the $l'=0$ term survives, so that

$$\lim_{r'\to\infty}\langle r|G^+(E)|r'\rangle = e^{ikr'}\times\text{function}(r). \quad (2.13b)$$

One recognizes that Eqs. (2.13a) and (2.13b) are the correct forms.[15]

Equation (2.12) has the same form as the complex basis function method discussed in Sec. I, i.e., with scalar product

$$\langle u_l^*|E-H|u_{l'}\rangle = \int_0^\infty dr\, u_l(r)(E-H)u_{l'}(r), \quad (2.14a)$$

$$\langle u_{l'}^*|g\rangle = \int_0^\infty dr\, u_{l'}(r)g(r), \quad (2.14b)$$

in which the function on the left-hand side of the integral is not complex conjugated. The important difference is that since the basis function $u_0(r)$ imposes the correct boundary condition on the vibrational trial functions, it is not necessary to invoke complex scaling/coordinate rotation or analytic continuation in order for all the matrix elements to exist. The matrix elements $\langle u_l|E-H|u_{l'}\rangle$ are obviously well behaved for $l,l'>0$, but they are also for $l$ or $l'=0$. To see this, consider the quantity

$$(H-E)u_0(r)$$

for $u_0(r)$ chosen as Eq. (2.10a), say. Since

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dr^2} + V(r),$$

and

$$E = \frac{\hbar^2k^2}{2m},$$

it is easy to show that

$$(H-E)u_0(r) = V(r)u_0(r) - \frac{\hbar^2\alpha}{2m}(2ik-\alpha)e^{ikr}e^{-\alpha r}.$$

So long as $V(r)\to 0$ faster than $1/r$, therefore, all the integrals

$$\int_0^\infty dr\, u_l(r)(H-E)u_0(r)$$

are well behaved, for $l=0$ as well as for $l>0$. The essential reason for this is that $u_0(r)$ satisfies the correct boundary condition as $r\to\infty$, so that

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dr^2} - E\right)u_0(r) = 0$$

as $r\to\infty$. (The reader may recognize the analogy of the present approach to the basis set calculation[16] of Siegert eigenvalues, for which a basis function that enforces the outgoing wave boundary condition is also included.)

In light of the above discussion it is clear that the present approach applies to long-range potentials, i.e., $V(r)\to r^{-n}$ as $r\to\infty$, so long as $n>1$. The complex scaling methods require that $V(r)$ vanish exponentially as $r\to\infty$.[4,5]

We note that Eq. (2.12) can be used to calculate any quantities[6] that can be expressed in terms of the Green"s function for a given Hamiltonian, but our present interest is in the application to the scattering operator $T = V + VG^+(E)V$. If the free particle wave function $\phi(r)$ is normalized as

$$\phi(r) = \sin(kr)/v^{1/2},$$

where $v=\hbar k/m$, then with Eq. (2.12) the $(1\times 1)$ $S$ matrix is

$$S-1 \equiv -(4i/\hbar)\langle\phi|T|\phi\rangle = -(4i/\hbar)\Big[\langle\phi|V|\phi\rangle + \sum_{l,l'=0}^{N}\langle\phi|V|u_l\rangle \times(\langle u_l^*|E-H|u_{l'}\rangle)^{-1}\langle u_{l'}^*|V|\phi\rangle\Big], \quad (2.15)$$

where $u_0(r)$ is the "special" basis function, Eq. (2.10). In order to separate the real and complex arithmetic in Eq. (2.15) as much as possible, it is useful to partition the $(N+1)\times(N+1)$ matrix

$$M_{l,l'} \equiv \langle u_l^*|E-H|u_{l'}\rangle \quad (2.16)$$

into the $1\times 1$ $l=l'=0$ block and the (real) $N\times N$ $l$, $l'=1,\ldots,N$ block. (This is an example of the standard $P$–$Q$ partitioning scheme.) Equation (2.15) can then be written in the matrix form

$$S = 1-(4i/\hbar)[V_{00} + \mathbf{V}^{\text{tr}}\cdot\mathbf{M}^{-1}\cdot\mathbf{V} + (V_0 - \mathbf{V}^{\text{tr}}\cdot\mathbf{M}^{-1}\cdot\mathbf{M}_0)\cdot(M_{00} - \mathbf{M}_0^{\text{tr}}\cdot\mathbf{M}^{-1}\cdot\mathbf{M}_0)^{-1}\cdot(V_0 - \mathbf{M}_0^{\text{tr}}\cdot\mathbf{M}^{-1}\cdot\mathbf{V})], \quad (2.17)$$

where $\mathbf{M}$ is the $N\times N$ $l,l'=1,\ldots,N$ (real) block of the matrix (2.16), $\mathbf{M}_0$ is the $N\times 1$ rectangular matrix $\langle u_l^*|E-H|u_0\rangle$, $l=1,\ldots,N$, $\mathbf{V}$ is $N\times 1$ matrix $\langle u_l^*|V|\phi\rangle$, $l=1,\ldots,N$, $V_0=\langle u_0^*|V|\phi\rangle$, $V_{00}=\langle\phi|V|\phi\rangle$, and "tr" denotes transpose (not adjoint, i.e., no complex conjugate). If one plans to carry out the calculation at many energies, then it may be useful to diagonalize the $N\times N$ Hamiltonian matrix $\langle u_l|H|u_{l'}\rangle$ so as to simplify the matrix inverse $\mathbf{M}^{-1}$ in Eq. (2.17).

Finally, we note that it is also possible to use Eq. (2.12) within a distorted wave representative for the scattering amplitude. Thus, if the potential $V=V_0+V_1$ and $\phi^{(\pm)}(r)$ are



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the scattering solutions for $V_0$ with boundary conditions

$$\phi^{(\pm)}(r) \sim e^{\pm i\eta_0} \sin(kr+\eta_0)/v^{1/2},$$

where $\eta_0$ is the phase shift for $V_0$, then Eq. (2.15) is modified as follows:

$$S = S_0 - (4i/\hbar)\Big[\langle\phi^{(-)}|V_1|\phi^{(+)}\rangle + \sum_{l,l'=0}^{N} \langle\phi^{(-)}|V_1|u_l\rangle \times (\langle u_l^*|E-H|u_{l'}\rangle)^{-1}\langle u_{l'}|V_1|\phi^{(+)}\rangle\Big], \tag{2.18}$$

where $S_0 = e^{2i\eta_0}$. The advantage of using a distorted wave representative is that the residual potential $V_1$ may be smaller and of shorter range than the total potential $V$, so that the basis $\{u_l\}$ has less work to do. This is a particularly useful possibility for inelastic and reactive scattering (see Sec. III). We note that it is also possible to build in distorted wavelike information by choice of the basis function $u_0(r)$. Thus one may choose

$$u_0(r) = e_0(r)(1-e^{-\alpha r}), \tag{2.19a}$$

where $e_0(r)$ is the (irregular) scattering wave function from $V_0(r)$ with asymptotic boundary condition

$$e_0(r) \sim e^{ikr}. \tag{2.19b}$$

Finally, we compare the result of the present Kohn-like expression for the $S$ matrix, i.e., Eq. (2.17), with a corresponding Kohn expression for the $K$ matrix. To make the correspondence as close as possible, we choose the complex function $u_0(r)$ to be that given by Eq. (2.10c),

$$u_0(r) = \tilde{u}_0(r) + i\sqrt{v}\,\phi(r), \tag{2.20a}$$

where

$$\tilde{u}_0(r) = \cos(kr)(1-e^{-\alpha r}), \tag{2.20b}$$

and as before,

$$\phi(r) = \sin(kr)/\sqrt{v}. \tag{2.20c}$$

One can then readily show that the various quantities in Eq. (2.17) can be expressed as

$$V_0 = \tilde{V}_0 + i\sqrt{v}\,V_{00}, \tag{2.21a}$$

$$\mathbf{M}_0 = \tilde{\mathbf{M}}_0 - i\sqrt{v}\,\mathbf{V}, \tag{2.21b}$$

$$M_{0,0} = \tilde{M}_{0,0} - 2i\sqrt{v}\,\tilde{V}_0 + vV_{0,0} - i\frac{\hbar v}{2}, \tag{2.21c}$$

where $\tilde{V}_0$, $\tilde{\mathbf{M}}_0$, and $\tilde{M}_{0,0}$ are the same quantities as $V_0$, $\mathbf{M}_0$, and $M_{0,0}$ with $u_0(r)$ replaced by $\tilde{u}_0(r)$. To facilitate the comparison it is useful to define the quantities

$$A = V_{00} + \mathbf{V}^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\mathbf{V}, \tag{2.22a}$$

$$B = \tilde{V}_0 - \mathbf{V}^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\tilde{\mathbf{M}}_0, \tag{2.22b}$$

$$C = \tilde{M}_{0,0} - \tilde{\mathbf{M}}_0^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\tilde{\mathbf{M}}_0. \tag{2.22c}$$

In terms of these quantities Eq. (2.17) for the $S$ matrix becomes

$$S-1 = -(4i/\hbar)\Big(AC + B^2 - i\frac{\hbar v}{2}A\Big)\Big/\Big(C + vA - 2i\sqrt{v}B - i\frac{\hbar v}{2}\Big). \tag{2.23}$$

For comparison, it is not hard to show that the unusual Kohn expression for the $K$ matrix is

$$K = -(2/\hbar)(A + B^2/C). \tag{2.24}$$

If one then uses the exact relation between $S$ and $K$,

$$S-1 = +2iK/(1-iK) \tag{2.25}$$

to obtain an $S$ matrix from Eq. (2.24), this gives

$$(S-1)_{\rm Kohn} = -(4i/\hbar)(AC+B^2)\Big/\Big(C + \frac{2i}{\hbar}AC + \frac{2i}{\hbar}B^2\Big). \tag{2.26}$$

Equations (2.26) and (2.23) do have much in common, but they are definitely not identical. Thus the usual Kohn expression for the $K$ matrix, used in Eq. (2.25) to obtain an $S$ matrix, is *not* equivalent to the direct use of the Kohn-like expression for the $S$ matrix.

## III. GENERALIZATION TO MULTICHANNEL SCATTERING

### A. Nonreactive scattering

Consider first nonreactive scattering. The notation will apply explicitly to collinear $A + BC(n_1) \to A + BC(n_2)$ vibrational excitation, but it should be clear that resulting expressions are generally applicable. The Hamiltonian has the form

$$H = \frac{-\hbar^2}{2\mu}\frac{d^2}{dR^2} + h + V(r,R), \tag{3.1}$$

where $h$ is the vibrational Hamiltonian for diatom BC, $r$ is the vibrational coordinate, and $R$ the translation coordinate. If one normalizes the plane wave state as

$$\Phi_{n_1}(r,R) \equiv \phi_{n_1}(r)\sin(k_{n_1}R)/v_{n_1}^{1/2}, \tag{3.2}$$

where $\{\phi_n(r)\}$ are the vibrational eigenstate of BC, $k_{n_1} = \sqrt{2\mu(E-\epsilon_{n_1})/\hbar^2}$, $v_{n_1} = \hbar k_{n_1}/\mu$, $E$ the total energy, and $\{\epsilon_n\}$ the vibrational eigenvalues, then the $S$ matrix for the $n_1 \to n_2$ transition is

$$S_{n_2,n_1} - \delta_{n_2,n_1} = -(4i/\hbar)[\langle\Phi_{n_2}|V|\Phi_{n_1}\rangle + \langle\Phi_{n_2}|VG^+(E)V|\Phi_{n_1}\rangle]. \tag{3.3}$$

Matrix elements of $G^+(E)$ are constructed as in Sec. II, e.g., Eqs. (2.1)–(2.6) apply as written. The wave function $\chi(r,R)$ corresponding to $|\chi\rangle$ of Eq. (2.5) now, however, is in the composite space $(r,R)$; it still satisfies the regularity boundary condition at $R=0$, and for $R\to\infty$ there are outgoing waves in all open vibrational channels

$$\chi(r,R) \sim \sum_{n=0}^{n_{\max}} \phi_n(r)e^{ik_nR}C_n, \tag{3.4}$$

where $n_{\max}$ is the vibrational quantum number of the highest open channel, and $\{C_n\}$ some constants. $\tilde{\chi}(r,R)$ is analogous. A linear variational trial function for $\chi(r,R)$ can thus be chosen as

$$\chi(r,R) = \sum_{l=0}\sum_n u_l^n(R)\phi_n(r)C_{ln}, \tag{3.5}$$

where the $l=0$ function $u_0^n(R)$ is chosen to impose the boundary condition



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$$u_0^n(R) \sim e^{ik_n R}. \tag{3.6a}$$

Possible specific choices are analogous Eq. (2.10);

$$u_0^n(R) = e^{ik_n R}(1 - e^{-\alpha R}), \tag{3.6b}$$

etc. The other translational functions $u_l^n(R)$ need not depend on the vibrational quantum number $n$, but may (i.e., one can use a different translational basis for each vibrational state, but this is not necessary). $\tilde{\chi}(r,R)$ is similarly expanded in this basis, and the linear expansion coefficients of $\chi$ and $\tilde{\chi}$ varied to obtain the generalization of Eq. (2.12),

$$\langle f|G^+|g\rangle = \sum_{l,n}\sum_{l'n'} \langle f|u_l^n \phi_n\rangle \times (\langle u_l^* \phi_n|E - H|u_{l'}\phi_{n'}\rangle)^{-1}\langle u_{l'}^* \phi_{n'}|g\rangle. \tag{3.7}$$

For $l = 0$ the sum over $n$ needs to include only the open vibrational states, but for $l > 0$ both open and closed channels must be included. Also, for $l > 0$ the functions $\{\phi_n(r)\}$ actually need not be the vibrational eigenfunctions—any complete set of functions in the $r$ degree of freedom is possible.

Using Eq. (3.7) in (3.3) thus gives the basis set approximation for the $S$ matrix as

$$S_{n_2,n_1} = \delta_{n_2,n_1} - (4i/\hbar)\Big[\langle\Phi_{n_2}|V|\Phi_{n_1}\rangle + \sum_{ln}\sum_{l'n'}\langle\Phi_{n_2}|V|u_l\phi_n\rangle(\langle u_l^*\phi_n|E - H|u_{l'}\phi_{n'}\rangle)^{-1} \times \langle u_{l'}^*\phi_{n'}|V|\Phi_{n_1}\rangle\Big]. \tag{3.8}$$

A distorted wave version may be useful here in order to simplify the task required of the basis set. Thus let $V(r,R) = V_0(R) + V_1(r,R)$, where $V_0$ is a distorting potential that causes only elastic scattering. The functions $\{\Phi_n(r,R)\}$ are now defined as

$$\Phi_n(r,R) = \phi_n(r) f_n(R), \tag{3.9a}$$

where $f_n(R)$ is the elastic scattering function from the potential $V_0(R)$ for energy $E_n = E - \epsilon_n$, normalized as

$$f_n(R) \sim \sin(k_n R + \delta_n)/v_n^{1/2}. \tag{3.9b}$$

The distorted wave expression for the $S$ matrix is then

$$S_{n_2,n_1} = S^{(0)}_{n_2,n_1} - (4i/\hbar)e^{i\delta_{n_1}}e^{i\delta_{n_2}}\Big[\langle\Phi_{n_2}|V_1|\Phi_{n_1}\rangle + \sum_{nl}\sum_{n'l'}\langle\Phi_{n_2}|V_1|u_l\phi_n\rangle(\langle u_l^*\phi_n|E - H|u_{l'}\phi_{n'}\rangle)^{-1} \cdot\langle u_{l'}^*\phi_{n'}|V_1|\Phi_{n_1}\rangle\Big], \tag{3.9c}$$

where

$$S^{(0)}_{n_2,n_1} = \delta_{n_2,n_1} e^{2i\delta_{n_1}}. \tag{3.9d}$$

Also, for either the plane wave [Eq. (3.8)] or the distorted wave [Eq. (3.9c)] version, the inverse of the matrix of $(E - H)$ can be carried out by partitioning the $l = 0$ block from the $l > 0$ block, a generalization of Eq. (2.17). The result is a straightforward matrix extension of Eq. (2.17); for the distorted wave version, for example, this gives

$$\mathbf{S} = \mathbf{S}_0 - (4i/\hbar)\mathbf{S}_0^{1/2}\cdot[\mathbf{V}_{00} + \mathbf{V}^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\mathbf{V} + (\mathbf{V}_0^{\rm tr} - \mathbf{V}^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\mathbf{M}_0)\cdot(\mathbf{M}_{00} - \mathbf{M}_0^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\mathbf{M}_0)^{-1} \cdot(\mathbf{V}_0 - \mathbf{M}_0^{\rm tr}\cdot\mathbf{M}^{-1}\cdot\mathbf{V})]\cdot\mathbf{S}_0^{1/2}, \tag{3.10}$$

where $\mathbf{V}_{00}$, $\mathbf{V}_0$, $\mathbf{M}_{00}$ (and $\mathbf{S}_0$) are "small" square matrices (dimension = number of open variational channels)

$$(\mathbf{V}_{00})_{n',n} = \langle\Phi_{n'}|V_1|\Phi_n\rangle, \tag{3.11a}$$

$$(\mathbf{V}_0)_{n',n} = \langle u_0^*\phi_{n'}|V_1|\Phi_n\rangle, \tag{3.11b}$$

$$(\mathbf{M}_{00})_{n',n} = \langle u_0^*\phi_{n'}|E - H|u_0\phi_n\rangle, \tag{3.11c}$$

$$(\mathbf{S}_0)_{n',n} = e^{2i\delta_n}\delta_{n',n}. \tag{3.11d}$$

$\mathbf{M}$ is the "large" square matrix

$$(\mathbf{M})_{l'n',ln} = \langle u_{l'}^*\phi_{n'}|E - H|u_l\phi_n\rangle, \tag{3.12}$$

and $\mathbf{V}$ and $\mathbf{M}_0$ are the rectangular matrices:

$$(\mathbf{V})_{l'n',n} = \langle u_{l'}^*\phi_{n'}|V_1|\Phi_n\rangle, \tag{3.13a}$$

$$(\mathbf{M}_0)_{l'n',n} = \langle u_{l'}^*\phi_{n'}|E - H|u_0\phi_n\rangle, \tag{3.13b}$$

where $l$ and $l' > 0$ for Eqs. (3.12) and (3.13). The only large matrix to be invented now is the real matrix $\mathbf{M}$, and as noted before, it may be useful to diagonalize $H_{n'l',nl}$, $l > 0$, once so that $\mathbf{M}^{-1}$ can be obtained easily for many different energies.

### B. Reactive scattering

The notation here will apply explicitly to a collinear $A + BC \rightarrow AB + C$ reaction. The plane wave representative for the reactive $S$ matrix is generalized from Eq. (3.3) by addition of an arrangement index $\gamma = a(A + BC)$, $b$ $(B + AC)$, or $c$ $(C + AB)$;

$$S_{\gamma_2 n_2,\gamma_1 n_1} - \delta_{\gamma_2\gamma_1}\delta_{n_2 n_1} = -(4i/\hbar)[\langle\Phi_{\gamma_2 n_2}|V_{\gamma_1}|\Phi_{\gamma_1 n_1}\rangle + \langle\Phi_{\gamma_2 n_2}|V_{\gamma_2}G^+V_{\gamma_1}|\Phi_{\gamma_1 n_1}\rangle], \tag{3.14}$$

where

$$\Phi_{\gamma n}(R_\gamma, r_\gamma) = \phi_n^\gamma(r_\gamma)\sin(k_n^\gamma R_\gamma)/v_{n_\gamma}^{1/2}. \tag{3.15}$$

Here $(r_\gamma, R_\gamma)$ are the vibrational and translational coordinates for arrangement $\gamma$, and $V_\gamma$ is the interaction potential for arrangement $\gamma[V_\gamma = V - v_\gamma(r_\gamma)$, where $v_\gamma$ is the vibrational potential for the diatom of arrangement $\gamma]$. (The reader should see Ref. 14, e.g., for a more complete discussion of the various coordinates for the different arrangements in reactive scattering.)

The primary new feature here is that the generic trial function $\chi \equiv G^+|\rangle$ has outgoing radial waves in all open channels of *all arrangements*. The asymptotic form of $\chi$ is thus

$$\chi \sim \sum_{n,\gamma}\phi_n^\gamma(r_\gamma)e^{ik_n^\gamma R_\gamma}C_{n\gamma}, \tag{3.16}$$

where the sum over $n$ is, for each $\gamma$, over all the open vibrational channels of that arrangement. $\chi$ may thus be expanded as

$$\chi = \sum_{l=0}\sum_{n,\gamma} u_l^\gamma(R_\gamma)\phi_n^\gamma(r_\gamma), \tag{3.17}$$

where $u_0^{\gamma n}(R_\gamma)$ is that of Eq. (3.6b), for example.

The treatment now follows precisely as that of Sec. III A, the only difference being that the vibrational quantum



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TABLE I. Fractional error in tan $\eta$ as a function of the number of basis functions; $k = 0.15$.[a]

| $N$ | SVP[b] | ROVP[c] | Present[d] |
|---|---|---|---|
| | | $\alpha = 2.5$ | |
| 2 | −1.6463 | 0.0423 | 0.4272 |
| 3 | 2.2367 | 0.0003 | 0.0270 |
| 4 | 0.2585 | 0.0002 | 0.0024 |
| 6 | 0.0164 | 0.0000 | 0.0000 |
| 8 | 0.0016 | 0.0000 | 0.0000 |
| 10 | 0.0013 | 0.0000 | 0.0000 |
| 12 | 0.0010 | 0.0000 | 0.0000 |
| | | $\alpha = 1.5$ | |
| 2 | 0.5974 | 0.0006 | 0.0006 |
| 3 | 0.0565 | 0.0000 | 0.0003 |
| 4 | 0.0068 | 0.0000 | 0.0000 |
| 6 | 0.0016 | 0.0000 | 0.0000 |
| 8 | 0.0009 | 0.0000 | 0.0000 |
| 10 | 0.0002 | 0.0000 | 0.0000 |
| 12 | 0.0000 | 0.0000 | 0.0000 |

[a] For the potential of Eq. (4.1), the model problem considered in Ref. 17.
[b] Results of the Schwinger variational principle, taken from Ref. 17.
[c] Results of the reactance operator variational principle, taken from Ref. 17.
[d] Our results from Eq. (2.15).

TABLE II. Fraction error in tan $\eta$ as a function of the number of basis functions; $k = 0.55$.[a]

| $N$ | SVP[b] | ROVP[c] | Present[d] |
|---|---|---|---|
| | | $\alpha = 2.5$ | |
| 2 | −0.3937 | 0.0123 | −0.0639 |
| 3 | −0.1236 | −0.0384 | −0.0104 |
| 4 | −0.0736 | −0.0043 | −0.0019 |
| 6 | −0.0631 | −0.0001 | −0.0003 |
| 8 | −0.0225 | 0.0000 | 0.0000 |
| 10 | −0.0031 | 0.0000 | 0.0000 |
| 12 | −0.0004 | 0.0000 | 0.0000 |
| | | $\alpha = 1.5$ | |
| 2 | −0.0985 | −0.0205 | −0.0081 |
| 3 | −0.0888 | 0.0000 | −0.0043 |
| 4 | −0.0649 | 0.0000 | 0.0000 |
| 6 | −0.0065 | 0.0000 | 0.0000 |
| 8 | −0.0005 | 0.0000 | 0.0000 |
| 10 | −0.0003 | 0.0000 | 0.0000 |
| 12 | 0.0000 | 0.0000 | 0.0000 |

[a] See footnote a of Table I.
[b] See footnote b of Table I.
[c] See footnote c of Table I.
[d] See footnote d of Table I.

number $n$ is augmented by the arrangement label $\gamma$. Both the plane wave [Eq. (3.8)] and distorted wave [Eqs. (3.9)] expressions, as well as the partitioned version [Eqs. (3.10)–(3.13)], apply as written with the modification $n \to \gamma, n$.

A final feature that arises here is the possibility of using a *multichannel* distorted wave representation, as in Ref. 14.

## IV. EXAMPLE

In this section we give the results of the present basis set method applied to two model problems, the first one elastic scattering and the second one inelastic scattering.

### A. Elastic scattering

The specific test problem here is that used recently by Staszewska and Truhlar (ST)[17] to test a scattering method they have proposed. It is a model for $s$-wave electron–atom scattering, the potential (in atomic units $m = \hbar = 1$) being

$$V(r) = -e^{-r}, \tag{4.1}$$

and we use the same basis set as ST,

$$u_l(r) = r^l e^{-\alpha r}, \quad l = 1,\ldots,N-1, \tag{4.2a}$$

and take

$$u_0(r) = e^{ikr}(1 - e^{-\alpha r}). \tag{4.2b}$$

Tables I and II give the fractional error in tan $\eta$,

$$\text{fractional error} = [(\tan\eta)_{\text{approximate}} - (\tan\pi_{\text{exact}}]/(\tan\eta)_{\text{exact}}, \tag{4.3}$$

for two different energies and various values of the nonlinear parameter $\alpha$ and numbers of basis functions $N$; SVP refers to the Schwinger variational principle (results from Ref. 17), ROVP to the reactance operator variational principle of ST (which can be shown to be the standard Kohn result with basis functions $\{G_0 u_l\}$), and "present" to the results given by Eq. (2.15). One sees that the present method gives results which are consistently better than the SVP and comparable to the ROVP method of Staszewska and Truhlar.

These results are very encouraging for the present method, for as emphasized above and unlike both the SVP and ROVP methods, it does not require construction of a zeroth order Green's function. That it does as well as these methods which do utilize $G_0$ is thus a very positive feature.

### B. Inelastic scattering

The real interest in any basis set method is for application to inelastic (or even more, to *reactive*) scattering, and our specific interest is to heavy particle, rather than electron scattering. Here we thus consider a standard benchmark for this topic, the Secrest–Johnson[18] model for collinear He + $H_2$ vibrationally inelastic collisions. All quantities are given in terms of the reduced units of Secrest and Johnson, to which we also refer the reader for the specific form of the potential energy function. The parameters $m = 0.667$ and $\gamma = 0.3$.

For this example we choose the translational basis functions $\{u_l(R)\}$ to be equally spaced floating Gaussians,

$$u_l(R) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-(\alpha/2)(R-R_l)^2}, \tag{4.4}$$

$l = 1,2,\ldots$, that have been extensively utilized recently by Light and co-workers.[19] The "special' translational function $u_0(R)$ is taken as

$$u_0(R) = e^{ikR} h(R), \tag{4.5a}$$

where the "cutoff" function $h(R)$ is

$$h(R) = \tfrac{1}{2}\{1 + \tanh[\alpha_1(R - R_1)]\}. \tag{4.5b}$$

The basis functions $\{\phi_n(r)\}$ for the vibrational degree of freedom are the harmonic oscillator eigenfunctions of $H_2$.



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TABLE III. Dependence of the $P_{01}$, $P_{02}$, and $P_{12}$ probabilities upon the real basis set parameter $\alpha$. $E = 6.0$, $\gamma = 0.315$, $m = 0.667$, $V_0 = 12$, $\alpha_1 = 0.5$, $R_1 = 10.0$, $N$ = total number of basis functions (real + complex).

| $N$ | $\alpha$ | $P_{01}$ | $P_{02}$ | $P_{12}$ |
|---|---|---|---|---|
| 15 | 1.62 | 0.281 58( − 1) | 0.125 39( − 4) | 0.145 84( − 2) |
| 15 | 2.77 | 0.286 31( − 1) | 0.122 40( − 4) | 0.170 29( − 2) |
| 15 | 3.87 | 0.282 72( − 1) | 0.610 85( − 5) | 0.153 14( − 2) |
| 18 | 1.78 | 0.275 08( − 1) | 0.123 99( − 4) | 0.147 81( − 2) |
| 18 | 1.97 | 0.319 72( − 1) | 0.129 21( − 4) | 0.147 19( − 2) |
| 18 | 2.19 | 0.290 40( − 1) | 0.120 17( − 4) | 0.146 93( − 2) |
| 18 | 2.46 | 0.303 07( − 1) | 0.120 33( − 4) | 0.147 32( − 2) |
| 18 | 2.77 | 0.298 49( − 1) | 0.117 00( − 4) | 0.147 16( − 2) |
| 18 | 3.62 | 0.299 73( − 1) | 0.164 47( − 4) | 0.151 75( − 2) |
| 18 | 4.20 | 0.297 07( − 1) | 0.871 70( − 5) | 0.150 68( − 2) |
| ∞[a] | | 0.300 58( − 1) | 0.112 73( − 4) | 0.147 23( − 2) |

[a] Results of numerical integration.

The methodology of Sec. III A has been applied in the distorted wave representation; i.e., Eq. (3.9). The translational functions $f_n(R)$ in Eq. (3.9) were determined by the elastic potential $V_0(R) = V_0 e^{-\gamma R}$. The first application is for $E = 6$, for which there are three open vibrational states; the $S$ matrix of Eq. (3.9c) is thus a $3\times3$ matrix. The sum over $n$ and $n'$ in Eq. (3.9c), however, must include closed as well as open vibrational states; we found that a total of five vibrational states in this expansion was sufficient to give converged results. For all the calculations reported in this section the results are quite insensitive to the parameters $\alpha_1$ and $R_1$ in cutoff function $h(R)$, Eq. (4.5b); we used the values $\alpha_1 = 0.5$, $R_1 = 10$.

Table III shows the inelastic transition probabilities $P_{01}(=P_{10})$, $P_{02}(=P_{20})$, and $P_{12}(=P_{21})$, for two different numbers of translational basis functions and for several values of the nonlinear parameter $\alpha$. Also given are the (presumably "exact") values obtained by numerical integration of the coupled-channel Schrödinger equation. One sees that the largest transition probabilities are reasonably well converged with 18 translational basis functions. The smallest probability, $P_{02}$, needs more basis functions for convergence (see below).

TABLE IV. Inelastic probabilities as a function of the total number of basis functions $N$. $R_{min} = 1.0$, $R_{max} = 18.0$; all other parameters as in Table III.

| $N$ | $\alpha$ | $P_{01}$ | $P_{02}$ | $P_{12}$ |
|---|---|---|---|---|
| 18 | 2.46 | 0.303 07( − 1) | 0.120 33( − 4) | 0.147 32( − 2) |
| 20 | 3.11 | 0.300 51( − 1) | 0.116 37( − 4) | 0.147 25( − 2) |
| 22 | 3.84 | 0.300 14( − 1) | 0.115 24( − 4) | 0.147 23( − 2) |
| 24 | 4.64 | 0.300 17( − 1) | 0.114 95( − 4) | 0.147 22( − 2) |
| ∞ | | 0.300 58( − 1) | 0.112 73( − 4) | 0.147 23( − 2) |

To illustrate the fact that the present method converges as the number of translational functions is increased, we choose the nonlinear parameter $\alpha$ [cf. Eq. (4.4)] so that the overlap of adjacent functions $\{u_l(R)\}$ is 0.5. The range $(R_{min}, R_{max})$ over which the Gaussians are distributed is fixed, so that increasing their number causes $\alpha$ to increase. Table IV shows that the three transition probabilities do indeed converge to their correct values as the number of translational functions is increased.

Table V gives similar results for the higher energy $E = 12$, for which there are six open vibrational states. The results for all the larger transition probabilities are again seen to be converged with 20–25 translational basis functions.

## V. CONCLUDING REMARKS

The purpose of this paper has been to point out a new kind of basis set method for treating quantum mechanical

TABLE V. Inelastic probabilities calculated in the distorted wave version. $E = 12.0$, $\gamma = 0.3$, $m = 0.667$, $V_0 = 20.0$, $\alpha_1 = 0.5$, $R_1 = 10.0$. Six vibrational channels of $H_2$ are open. Results are fully converged using eight vibrational basis functions.

| $N$ | $\alpha$ | $R_{max}$ | $P_{01}$ | $P_{14}$ | $P_{23}$ | $P_{25}$ |
|---|---|---|---|---|---|---|
| 15 | 2.77 | 14.0 | 0.703 55 | 0.394 88( − 3) | 0.216 36 | 0.535 37( − 5) |
| 15 | 2.39 | 15.0 | 0.490 20 | 0.374 75( − 3) | 0.250 17 | 0.541 59( − 5) |
| 15 | 2.08 | 16.0 | 0.456 15 | 0.878 69( − 3) | 0.151 08 | 0.302 14( − 5) |
| 15 | 1.83 | 17.0 | 0.547 08 | 0.365 43( − 2) | 0.417 43 | 0.623 41( − 5) |
| 15 | 1.63 | 18.0 | 0.603 67 | 0.778 33( − 2) | 0.617 20( − 1) | 0.433 60( − 5) |
| 20 | 5.32 | 14.0 | 0.393 15 | 0.298 16( − 3) | 0.223 37 | 0.709 39( − 5) |
| 20 | 4.58 | 15.0 | 0.387 21 | 0.343 45( − 3) | 0.227 86 | 0.980 18( − 5) |
| 20 | 3.99 | 16.0 | 0.381 51 | 0.371 12( − 3) | 0.229 25 | 0.253 88( − 5) |
| 20 | 3.51 | 17.0 | 0.471 21 | 0.410 35( − 3) | 0.230 58 | 0.203 84( − 5) |
| 20 | 3.11 | 18.0 | 0.238 57 | 0.393 39( − 3) | 0.229 73 | 0.254 33( − 5) |
| 20 | 2.49 | 20.0 | 0.624 19( − 1) | 0.244 61( − 3) | 0.232 90 | 0.201 37( − 5) |
| 25 | 8.68 | 14.0 | 0.394 15 | 0.306 21( − 3) | 0.223 24 | 0.347 08( − 4) |
| 25 | 7.48 | 15.0 | 0.393 41 | 0.343 51( − 3) | 0.228 57 | 0.138 03( − 4) |
| 25 | 6.52 | 16.0 | 0.392 86 | 0.350 24( − 3) | 0.230 23 | 0.165 72( − 5) |
| 25 | 5.73 | 17.0 | 0.392 40 | 0.378 37( − 3) | 0.231 24 | 0.209 68( − 5) |
| 25 | 5.08 | 18.0 | 0.393 45 | 0.398 35( − 3) | 0.231 83 | 0.264 91( − 5) |
| 25 | 4.53 | 19.0 | 0.386 82 | 0.401 46( − 3) | 0.231 76 | 0.206 92( − 5) |
| 25 | 4.06 | 20.0 | 0.395 57 | 0.408 56( − 3) | 0.231 30 | 0.194 64( − 5) |
| ∞[a] | | | 0.394 23 | 0.389 71( − 3) | 0.233 45 | 0.201 37( − 5) |

[a] Results of numerical integration.



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scattering problems. It is a Kohn-like variational expression applied directly to the $S$ (or $T$) matrix rather than to the $K$ matrix as is usually done; the result obtained is not equivalent to the usual Kohn result and is, in fact, considerably preferable to it. The resulting expression is seen to have a similar structure to that of the complex scaling methods, but is preferable to them, especially for large values of the wave vector $k$ (i.e., for heavy particle scattering). Its primary advantage over the Schwinger variational method is that the zeroth order Green's function $G_0$ is not required for its application.

Application to some simple model problems of elastic and inelastic scattering show that the method is well behaved with respect to increasing the size of the basis set and to the variation of nonlinear parameters in the basis functions. In the area of heavy particle scattering its potentially most significant contribution is to reactive scattering, and this work is in progress. It should also be of use in electron–atom/molecule scattering which also involves exchange processes. It also seems clear that considerable improvement in the method can be made by undertaking systematic studies to see how to tune the basis set to the specific problem at hand.

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